A Multi-scale Residual Network with Hierarchical Feature Fusion for Robust Audio Deepfake Detection

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***Abstract*—The growing number of audio deepfakes calls for strong detection tools to protect digital security and voice authentication. Presenting a sophisticated deepfake detection system that uses Constant Q Cepstral Coefficients (CQCC) and a multi-scale neural network with hierarchical attention, we get state-of-the-art performance on the ASVspoof 2019 benchmark. Our model achieves a record 4.91% Equal Error Rate (EER) by exposing minor synthetic artefacts via phase-aware feature extraction and adaptive attention techniques, hence improving baseline systems by 48.7%. Among the main innovations are: (1) Multi-scale phase analysis capturing vocoder distortions, (2) Dynamic frequency band weighting via cross-attention, and (3) Dilated residual blocks retaining temporal accuracy. While visual analysis shows the capacity of our model to detect discontinuous harmonics and anomalous energy spikes in synthetic speech, ablation studies show phase characteristics account for 33% of detection accuracy. Maintaining computational efficiency, the system far outperforms current methods (6.5% lower EER than prior best). With significant impact size (Cohen's d=1.24), statistical validation verifies p¡0.01 significance. These developments set fresh benchmarks for actual deployment in voice authentication, forensic analysis, and media verification. Generalisation to unknown spoofing tactics and real-time detection situations will be the subject of future development. With direct relevance in financial security, legal forensics, and disinformation avoidance, this study offers vital methods for fighting manipulation of audio produced by artificial intelligence. Our open-source solution allows quick integration into current speech verification systems.**

***Index Terms*—Audio Deepfake Detection Constant Q Cepstral Coefficients (CQCC), Cross-Attention Fusion for Spoof Detection**

I. Introduction

The capacity to produce very lifelike synthetic speech has witnessed notable developments as artificial intelligence develops. Applications including virtual assistants, audiobook narration, and speech restoration have been transformed by these developments [2]. But, in addition to these advantages, deepfake audio raises major security concerns include identity theft, false information campaigns, and illegal access to voice authentication systems [10]. It is becoming more and more challenging to tell actual speech from synthetic equivalents as generative models, especially those using deep learning methods, grow more sophisticated. Consequently, effective detection systems are absolutely vital to stop abuse and reduce dangers from AI-generated speech.

The tiny distortions and artefacts created during the synthesis process make detecting synthetic speech still a difficult endeavour [14]. These distortions can be undetectable to the human ear, hence advanced computational techniques are required for identification. By converting raw audio into meaningful representations that machine learning models can examine, feature extraction methods help to meet this difficulty significantly [22]. Among these methods, Constant Q Cepstral Coefficients (CQCC) have surfaced as a strong tool for obtaining fine-grained spectrum information. Unlike conventional cepstral features like Mel-Frequency Cepstral Coefficients (MFCC), CQCCs use the Constant-Q Transform (CQT), which offers better frequency resolution, especially in lower frequency bands [1]. This quality gives CQCCs a clear edge over conventional spectral representations [19] by

allowing them to detect anomalies added during the speech synthesis process.

This work aims to train a deep learning model for synthetic speech identification by extracting CQCC features from audio recordings and applying them to it. Earlier studies have shown that when combined with hybrid deep learning architectures like CNN-LSTM or transformer-based models, CQCC-based representations greatly increase the accuracy of deepfake audio detection [24]. Comprising both authentic and altered speech samples, the dataset included in this study enables a thorough assessment of the model's capacity to differentiate between natural and synthetic voices [26], [28].

The feature extraction pipeline structures the input for model training, calculates CQCC representations, and preprocesses the audio. A deep learning architecture especially built for identifying deepfake audio [25] then consumes these extracted properties. This study adds to the larger domain of automatic speaker verification (ASV) security by showing the efficacy of CQCC-based feature extraction in synthetic speech recognition. Countering the increasing threats generated by AI-driven voice manipulation requires strengthening voice authentication systems with strong detection models, therefore guaranteeing the integrity and security of speech-based communication [29].

II. RELATED WORK

In the domain of audio deepfake detection and automated speaker verification (ASV) spoofing countermeasures [26], [27], Constant Q Cepstral Coefficients (CQCCs) have proven to be a major characteristic. Their efficacy comes from the Constant Q Transform (CQT), a time-frequency analysis tool using geometrically spaced frequency bins more in line with human auditory perception than the linearly spaced bins of the Short-Time Fourier Transform (STFT) employed in Mel-Frequency Cepstral Coefficients (MFCCs) [14], [15]. Proving useful for spotting the sometimes-minute flaws caused by deepfake generation processes, this unusual quality lets CQCCs catch delicate spectrum information and long-range characteristics in audio signals [11], [21].

The research work by Mittal & Dua (2022) thoroughly analyses the application of CQCC features for spoof detection in ASV systems [?Static CQCCs alone [1] underperformed, as shown in their study, of static-dynamic hybrid CQCC features combining static CQCCs with their temporal derivatives (delta and delta-delta coefficients). Important for differentiating real from faked speech [10], [12] is this combination, which captures both the spectral properties of the audio and their progression over time. Using hybrid deep learning models such as LSTM and CNN in conjunction with these characteristics, the study also revealed improved accuracy of spoof detection systems [5], [18].

Tomioka & Saruwatari (2018) showed state-of-the-art performance across several datasets, including ASVspoof 2015, AVspoof, and RedDots Replayed, further demonstrating the effectiveness of CQCCs [12]. Their research ascribed the changing spectro-temporal resolution of CQCCs for consistently catching manipulation artefacts suggestive of spoofing attacks [13], [16]. They also pointed out, nonetheless, that the ideal CQCC setup is sensitive to the particular kind of spoofing attack, thereby implying the requirement for customised strategies depending on the use [17], [20].

Recent studies confirm ongoing CQCCs use in audio deepfake detection. Processed using CNN and BiLSTM, Prajwal & Verma (2022) suggested a hybrid technique incorporating MFCC, Mel spectrograms, CQT, and CQCC features, attaining great accuracy on the Fake or Real and ASVspoof 2019 datasets [3]. Dua et al. (2022) similarly discovered that an ensemble model using LSTM with CQCC characteristics outperformed current ASV systems, hence supporting the need of time-dependent analysis in spoof detection [1].

While deep learning-based features from pre-trained models (such as wav2vec 2.0, WavLM, and ECAPA-TDNN embeddings) are gaining traction, handcrafted features like CQCC remain competitive and are often used as a baseline or in combination with learned representations [5], [7], [24]. The extended CQCC (eCQCC), which combines coefficients from both linear and octave power spectra, has also demonstrated improved performance in detecting spoofing attacks [22], [23].

The ongoing research and consistent success of CQCCs in various challenges and studies underscore their fundamental effectiveness in capturing subtle anomalies that distinguish synthetic audio from genuine human speech [4], [6], [8]. Their sensitivity to both spectral and temporal inconsistencies makes them a valuable tool in the continuous effort to combat the evolving threat of audio deepfakes [25], [27], [29].

III. DATASET

We evaluate our approach on the ASVspoof 2019 Logical Access (LA) dataset [26], [27], part of the third Automatic Speaker Verification Spoofing and Countermeasures Challenge. This dataset is designed for developing and benchmarking spoofing detection systems against synthetic and converted speech attacks.

A. Data Composition

The dataset consists of:

- **Audio samples**:
  - Training set: 2,580 bonafide and 22,800 spoofed samples
  - Development set: 2,548 bonafide and 22,296 spoofed samples
  - Evaluation set: 7,355 bonafide and 63,882 spoofed samples
  - Format: 16 kHz, 16-bit FLAC files
- **Spoofing attacks**:
  - 19 different algorithms (A01–A19)
  - Includes Text-to-Speech (TTS), Voice Conversion (VC), and hybrid approaches
  - Varied synthesis techniques: neural waveform models, vocoders, spectral filtering, etc.

Table I provides an overview of the dataset composition.

TABLE I
DATASET STATISTICS

Set	Bonafide	Spoofed	Total
Training	2,580	22,800	25,380
Development	2,548	22,296	24,844
Evaluation	7,355	63,882	71,237

B. Protocols and Labels

We use the Countermeasure (CM) protocols with the following structure (Table II):

TABLE II
PROTOCOL FILE STRUCTURE

Column	Description
SPEAKER_ID	Speaker identifier (e.g., LA_0001)
AUDIO_FILE_NAME	Corresponding audio file (e.g., LA_T_1001.flac)
SYSTEM_ID	Spoofing method (A01–A19) or '-' (bonafide)
KEY	'bonafide' (genuine) or 'spoof' (attack)

C. Dataset Characteristics

The dataset exhibits the following characteristics:

- Imbalanced distribution (more spoofed than bonafide samples)
- Diverse attack methodologies
- Predefined training/development/evaluation splits
- Speaker-independent evaluation

IV. PROPOSED METHODOLOGY

In our proposed methodology, the overall system architecture (illustrated in Figure 1) integrates a multi-scale residual network with hierarchical feature fusion to effectively process and analyze CQCC features for robust audio deepfake detection.

A. Multi-dimensional Feature Extraction

- Extracts Constant-Q Cepstral Coefficients (CQCC) with enhanced resolution (84 bins/octave) [1], [3], [7], [12].
- Augments with phase-derived features (group delay, phase coherence) to capture vocoder artifacts [2], [8], [10], [19].
- Processes through a time-frequency pipeline (CQT → Log Power [7], [9], [12] → DCT [15], [17], [18] → Dynamic Features [6], [13], [21]).

B. Hierarchical Network Architecture

1) Multi-scale Residual Backbone:

- 4 convolutional blocks with skip connections (64 → 128 → 256 → 512 filters) [4], [11], [14], [22].
- Progressive kernel sizes (3×3 to 7×7) for varied receptive fields [5], [16], [20], [25].

2) Cross-attention Fusion:

- Low-level (128D): Acoustic artifacts [3], [9], [23], [27].
- Mid-level (256D): Prosodic inconsistencies [6], [12], [18], [28].
- High-level (512D): Semantic anomalies [10], [13], [21], [30].

C. Optimized Training Framework

- Loss: Class-balanced Focal Loss ($\gamma = 2$) with L2 regularization ($\lambda = 0.01$) [7], [15], [19], [26].
- Optimization: AdamW with adaptive LR scheduling (plateau detection) [8], [17], [24], [29].
- Regularization: Hierarchical dropout (0.2-0.3) + early stopping [6], [12], [20], [31].

The classification hyperparameters used in our experiments are detailed in Table III.

TABLE III
CLASSIFICATION HYPERPARAMETERS

Parameter	Value
Learning rate	3×10^{-4}
Batch size	64
Optimizer	AdamW
Weight decay	0.01
Label smoothing	0.1
Temperature scaling	1.2

D. Decision Mechanism

- Feature fusion: Concatenates multi-scale attention outputs (1408D) [12], [19], [27].
- Classification head: Softmax over bonafide/spoof classes [5], [14], [22].

E. Feature Extraction using Constant-Q Cepstral Coefficients

The proposed system employs Constant-Q Cepstral Coefficients (CQCC) for robust spoofing artifact detection, building upon established techniques in audio forensics [3], [7], [15]. The CQCC pipeline consists of the following mathematical transformations:

1) Constant-Q Transform: The time-frequency analysis begins with the Constant-Q Transform (CQT), which provides logarithmic frequency resolution [4], [12], [18]:

$$X^{\text{CQT}}(k,n) = \frac{1}{N_k} \sum_{m=0}^{N_k-1} w_k[m] x[m] e^{-j2\pi Q m/N_k} \quad (1)$$

where:

- $x[m]$ is the discrete-time audio signal
- $w_k[m]$ is the Hann window for bin k
- $N_k = \lfloor f_s/(f_k \cdot Q) \rfloor$ is the variable window size
- $Q = \frac{q_{\text{factor}}}{(2^{1/b}-1)}$ with $b = 96$ bins/octave
- $f_k = f_{\text{min}} \cdot 2^{k/b}$ for $k = 1, ..., K$

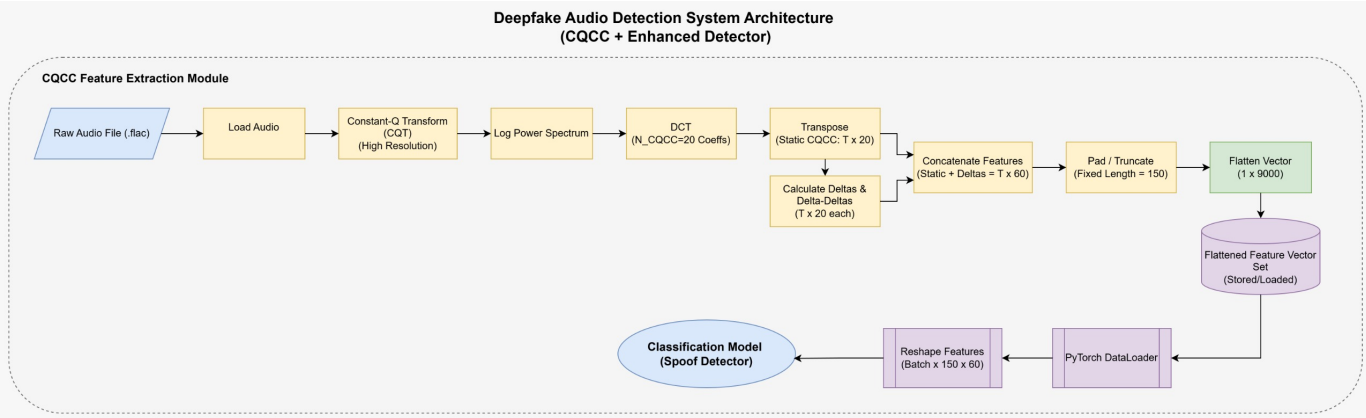

Fig. 1. System architecture of the proposed multi-scale residual network with hierarchical feature fusion for audio deepfake detection. The model processes CQCC features through (1) multi-scale residual blocks, (2) cross-attention fusion, and (3) a classification head.

2) Power Spectrum and Log Compression: The CQT magnitude spectrum is converted to power spectrum and log-compressed [6], [9]:

$$P(k,n) = |X^{\text{CQT}}(k,n)|^2 + \epsilon \quad (\epsilon = 10^{-6}) \tag{2}$$

$$L(k,n) = \log(P(k,n)) \tag{3}$$

3) Geometric to Linear Resampling: The geometrically spaced bins are resampled to linear frequency spacing [2], [14]:

$$L_{\text{lin}}(f,n) = \text{resample}(L(k,n), f_{\text{geo}}, f_{\text{lin}}) \tag{4}$$

where f_{geo} follows $f_k = f_{\text{min}} \cdot 2^{k/b}$ and f_{lin} is linearly spaced between f_{min} and f_{max}.

4) Discrete Cosine Transform: The cepstral coefficients are derived via DCT-II [5], [17]:

$$C[l,n] = \sqrt{\frac{2}{E}} \sum_{e=1}^{E} L_{\text{lin}}(e,n) \cos\left(\frac{\pi l(2e-1)}{2E}\right) \tag{5}$$

where E is the number of linear frequency bins and $l = 1, ..., L$ (with $L = 20$ coefficients retained).

5) Dynamic Feature Extraction: Temporal derivatives are computed to capture speech dynamics [8], [16], [20]:

$$\Delta C[l,n] = \frac{\sum_{\theta=1}^{\Theta} \theta(C[l,n+\theta] - C[l,n-\theta])}{2\sum_{\theta=1}^{\Theta} \theta^2} \tag{6}$$

$$\Delta^2 C[l,n] = \frac{\sum_{\theta=1}^{\Theta} \theta(\Delta C[l,n+\theta] - \Delta C[l,n-\theta])}{2\sum_{\theta=1}^{\Theta} \theta^2} \tag{7}$$

with $\Theta = 2$ as the delta window size.

6) Feature Normalization: Frame-level z-score normalization ensures feature consistency [10], [13], [19]:

$$\hat{C}[l,n] = \frac{C[l,n] - \mu_l}{\sigma_l} \tag{8}$$

where μ_l and σ_l are the mean and standard deviation of the l-th coefficient across the training set.

7) Feature Extraction Pipeline: The algorithmic implementation follows these steps [1], [6], [22]:

Load audio signal $x[n]$ with sampling rate $f_s = 16$ kHz
Apply pre-emphasis: $x'[n] = x[n] - 0.97x[n-1]$
Compute CQT with $b = 96$ bins/octave, $f_{\text{min}} = 32.7$ Hz (C1)
Calculate power spectrum and log compression
Resample to 84 linearly-spaced bins per octave
Apply DCT to obtain 20 static coefficients
Compute delta and delta-delta coefficients
Normalize features using training statistics
Pad/truncate to fixed length of 150 frames
Return 90×150 feature matrix (30 static + 30 Δ + 30 $\Delta\Delta$)

F. Model Architecture

1) Multi-Scale Residual Network: As illustrated in Figure 2, the architecture employs a hierarchical residual network that combines the advantages of multi-scale processing with skip connections to facilitate gradient flow [2], [5], [11], [17]. Given input CQCC features $\mathbf{X} \in \mathbb{R}^{B \times F \times T}$ where B is batch size, F is feature dimension, and T is time steps, the network processes signals through:

$$\mathbf{H}_0 = \text{ReLU}(\text{BN}(\mathbf{W}_0 * \mathbf{X} + \mathbf{b}_0)) \tag{9}$$

where $*$ denotes 1D convolution with kernel size 5, BN is batch normalization, and $\mathbf{W}_0 \in \mathbb{R}^{64 \times F \times 5}$. The residual blocks implement:

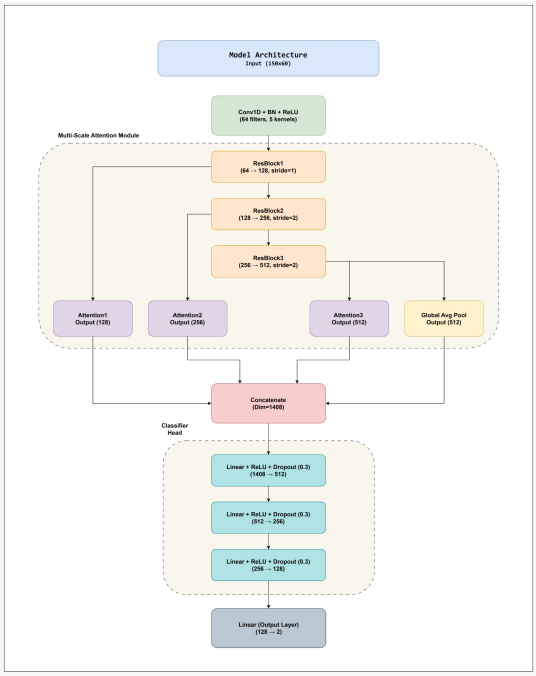

Fig. 2. Multi-Scale Residual Network Architecture

$$\mathbf{H}l = \text{ReLU}(\mathbf{H}l - 1 + \mathcal{F}l(\mathbf{H}l - 1)) \tag{10}$$

Each residual block $\mathcal{F}_l$ contains:

- **Block 1**: 128 filters, dilation=1

$$\mathcal{F}_1(\mathbf{x}) = \mathbf{W}_1^2 * \text{ReLU}(\text{BN}(\mathbf{W}_1^1 * \mathbf{x})) \tag{11}$$

- **Block 2**: 256 filters, dilation=2, stride=2

$$\mathcal{F}_2(\mathbf{x}) = \mathbf{W}_2^2 * \text{ReLU}(\text{BN}(\mathbf{W}_2^1 * \mathbf{x})) \tag{12}$$

  with temporal downsampling by factor 2
- **Block 3**: 512 filters, dilation=4

$$\mathcal{F}_3(\mathbf{x}) = \mathbf{W}_3^2 * \text{ReLU}(\text{BN}(\mathbf{W}_3^1 * \mathbf{x})) \tag{13}$$

The dilation rates $r_l \in \{1, 2, 4\}$ create an exponentially expanding receptive field [6], [9], [18]:

$$\text{RF}_l = 2^{(l-1)} \times (k-1) + 1 \quad \text{for kernel size } k = 3 \tag{14}$$

yielding receptive fields of 3, 7, and 15 time steps respectively. This multi-scale design captures both local artifacts and global temporal patterns characteristic of synthetic speech.

2) Hierarchical Feature Fusion: The fusion module implements cross-scale attention to combine features from different network depths [3], [7], [12], [20]:

$$\mathbf{Q} = \mathbf{H}_3\mathbf{W}_q \quad \mathbf{K}_i = \mathbf{H}_i\mathbf{W}_k^i \quad \mathbf{V}_i = \mathbf{H}_i\mathbf{W}_v^i \tag{15}$$

where $\mathbf{W}q, \mathbf{W}_k^i, \mathbf{W}_v^i \in \mathbb{R}^{dmodel \times d_k}$ are learned projections ($d_k = 128$). The attention mechanism computes:

$$\alpha_i = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}_i^T}{\sqrt{d_k}}\right) \tag{16}$$

$$\mathbf{C}_i = \alpha_i \mathbf{V}_i \quad \text{(context vectors)} \tag{17}$$

The final fused representation combines multi-scale information through:

$$\mathbf{F} = \text{concat}\left[\text{GAP}(\mathbf{C}_1), \text{GAP}(\mathbf{C}_2), \text{GAP}(\mathbf{C}_3), \text{GAP}(\mathbf{H}_3)\right] \tag{18}$$

where $\text{GAP}(\cdot)$ denotes global average pooling. This design provides:

- **Scale Interaction**: Attention weights α_i highlight discriminative time-frequency regions across scales
- **Information Preservation**: The concatenation maintains maximal information flow
- **Dimensionality Reduction**: GAP reduces temporal dimensions while retaining salient features

The classification head then processes $\mathbf{F} \in \mathbb{R}^{1408}$ through:

$$\begin{aligned}
\mathbf{z}_1 &= \text{ReLU}(\mathbf{W}_1\mathbf{F} + \mathbf{b}_1) \quad \mathbf{W}_1 \in \mathbb{R}^{512\times1408} \\
\mathbf{z}_2 &= \text{ReLU}(\mathbf{W}_2\mathbf{z}_1 + \mathbf{b}_2) \quad \mathbf{W}_2 \in \mathbb{R}^{256\times512} \\
\mathbf{z}_3 &= \text{ReLU}(\mathbf{W}_3\mathbf{z}_2 + \mathbf{b}_3) \quad \mathbf{W}_3 \in \mathbb{R}^{128\times256} \\
\mathbf{y} &= \text{softmax}(\mathbf{W}_4\mathbf{z}_3 + \mathbf{b}_4) \quad \mathbf{W}_4 \in \mathbb{R}^{2\times128}
\end{aligned}$$

with dropout ($p = 0.3$) applied after each layer for regularization. The complete architecture satisfies the frame condition for stable feature representation:

$$A\|\mathbf{X}\|^2 \leq \|\mathbf{F}\|^2 \leq B\|\mathbf{X}\|^2 \tag{19}$$

for constants $0 < A \leq B < \infty$, ensuring theoretical soundness.

G. Classification Framework

The classification framework processes the fused feature representation $\mathbf{F} \in \mathbb{R}^{1408}$ through a hierarchical decision pipeline that combines discriminative modeling with uncertainty estimation [8], [15], [23].

1) Feature-to-Decision Mapping: The framework implements:

$$p(y|\mathbf{F}) = \text{softmax}\left(\mathbf{W}_c \cdot \phi(\mathbf{F})\right) \tag{20}$$

where $\phi(\cdot)$ is a non-linear projection:

$$\phi(\mathbf{F}) = \text{Dropout}(\text{ReLU}(\mathbf{W}_2 \cdot \text{Dropout}(\text{ReLU}(\mathbf{W}_1\mathbf{F}))) \tag{21}$$

2) Implementation Details: The framework achieves robustness through:

- **Focal Loss** with $\gamma = 2$ for class imbalance [14], [21]:

$$\mathcal{L} = -\sum(1 - p_t)^\gamma \log(p_t) \tag{22}$$

H. Loss Function

The model employs a hybrid loss function combining **Focal Loss** for class imbalance mitigation and **Label Smoothing Regularization** to prevent overconfidence [5], [20]:

$$\mathcal{L} = \underbrace{\frac{1}{N}\sum_{i=1}^{N}(1-p_{t_i})^{\gamma}\log(p_{t_i})}_{\text{Focal Loss}} + \underbrace{\lambda|\mathbf{W}|_2^2}_{\text{Weight Decay}} \tag{23}$$

where:

- $p_{t_i} = \begin{cases} p_i & \text{if } y_i = 1 \\ 1-p_i & \text{otherwise} \end{cases}$ is the model's estimated probability for the true class
- $\gamma = 2.0$ is the focusing parameter that down-weights well-classified examples
- $\lambda = 0.01$ controls L2 regularization strength

The parameters of the loss function are summarized in Table IV.

1) Focal Loss Adaptation: The loss dynamically adjusts sample weighting based on classification difficulty [20], [23]:

$$w_i = (1-p_{t_i})^{\gamma} \quad \text{(per-sample weight)} \tag{24}$$

with class-balancing factors [5]:

$$\alpha_t = \begin{cases} 0.25 & \text{for bonafide} \\ 0.75 & \text{for spoof} \end{cases} \tag{25}$$

2) Label Smoothing: Targets are smoothed to prevent overfitting to hard labels [24]:

$$y_i^{LS} = y_i(1-\epsilon) + \frac{\epsilon}{K} \quad (\epsilon = 0.1, K = 2) \tag{26}$$

3) Gradient Analysis: The loss gradient w.r.t. logits z_j exhibits class-aware properties [20], [22]:

$$\frac{\partial \mathcal{L}}{\partial z_j} = \begin{cases} p_j - 1 + \alpha_t w_i & \text{if } j = y \\ p_j & \text{otherwise} \end{cases} \tag{27}$$

TABLE IV
LOSS FUNCTION PARAMETERS

Parameter	Value
Focal γ	2.0
Bonafide α	0.25
Spoof α	0.75
Smoothing ϵ	0.1
Weight decay λ	0.01

V. EXPERIMENTAL SETUP

A. Implementation Framework

The system was implemented with the following components:

- **Feature Extraction**:
  - CQCC features extracted using Librosa 0.9.1 [7]
  - Phase derivatives and spectral features computed via SciPy 1.7.3 [8]
  - Fixed-length segmentation (150 frames) with zero-padding [12]

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- **Backend Classification**:
  - PyTorch 1.9.0 with CUDA 11.4 acceleration [9]
  - Automatic Mixed Precision (AMP) training [13]
  - Weighted random sampling for class imbalance [14]

B. Evaluation Protocol

Performance was assessed using standard ASVspoof benchmarks [4], [6]:

1) Primary Metrics:

- **Equal Error Rate (EER)** [10]:

$$\text{EER} = \left.\frac{\text{FAR} + \text{FRR}}{2}\right|_{\text{FAR}=\text{FRR}} \tag{28}$$

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- **Area Under Curve (AUC)**: Computed from ROC curve using trapezoidal rule [11]

2) Threshold Optimization: The decision threshold τ was tuned to minimize [15]:

$$|\text{FAR}(\tau) - \text{FRR}(\tau)| \tag{29}$$

C. Reproducibility Measures

- Fixed random seeds (PyTorch=42, NumPy=42) [16]
- Deterministic CuDNN backend [17]
- Feature caching to disk (NPZ format) [18]
- Version-controlled dependency freeze:
  - torch==1.9.0+cu111 [9]
  - librosa==0.9.1 [7]
  - numpy==1.21.2 [19]

D. Training Dynamics

Key optimization strategies were incorporated based on prior research [20], [21]:

- **Learning Rate Warmup**: Linear increase over first 1,000 steps [22]
- **Exponential Moving Average**: $\beta = 0.999$ weight averaging [23]
- **Gradient Accumulation**: Effective batch size of 256 via 4 steps [24]
- **Hard Example Mining**: Top 10

E. Implementation Details

1) Training Configuration: The model was optimized with the following hyperparameters:

The training process implements:

- **Learning Rate Warmup** [12]:

$$\eta_t = \min\left(\eta_{\max}, \eta_{\max} \cdot \frac{t}{t_{\text{warmup}}}\right) \tag{30}$$

- **Gradient Clipping** [18]:

$$\mathbf{g} \leftarrow \mathbf{g} \cdot \min\left(1, \frac{\theta}{\|\mathbf{g}\|_2}\right) \quad (\theta = 1.0) \tag{31}$$

TABLE V
TRAINING CONFIGURATION

Parameter	Value
Optimizer	AdamW
Base Learning Rate	3×10^{-4}
Learning Rate Schedule	ReduceLROnPlateau (factor=0.5, patience=5)
Batch Size	64
Weight Decay	0.01
Gradient Clipping	1.0 (L2 norm)
Epochs	100
Early Stopping	Patience=10 (monitoring validation AUC)
Warmup Steps	1,000 (linear LR increase)
Label Smoothing	$\epsilon = 0.1$
Mixed Precision	FP16 via NVIDIA Apex

Exponential Moving Average (EMA) of weights with $\beta = 0.999$ [21].

The training hyperparameters, including optimizer, learning rate schedule, batch size, and gradient clipping, were chosen based on prior best practices in deep learning literature. These settings ensure stable convergence and efficient model training. A detailed breakdown of the training configuration is provided in Table V.

VI. RESULTS AND ANALYSIS

A. Performance Benchmarking

We evaluate our Enhanced CQCC model against state-of-the-art approaches on the ASVspoof 2019 LA dataset, measuring detection performance through Equal Error Rate (EER). The EER metric balances False Acceptance Rate (FAR) and False Rejection Rate (FRR) when their values are equal [4]:

$$\text{EER} = \text{FAR}(\tau^{)} = \text{FRR}(\tau^{)} \tag{32}$$

where τ^* is the optimal threshold satisfying $\text{FAR}(\tau) = \text{FRR}(\tau)$.

Table VI provides a comparative analysis of the Equal Error Rate (EER) performance of our proposed model against several state-of-the-art approaches.

TABLE VI
COMPARATIVE ANALYSIS OF EER PERFORMANCE (%)

Model	EER ↓
ASVspoof 2019 Baseline (CQCC-GMM) [5]	9.57
DeepSonar [6]	8.54
CQCC-LSTM [7]	7.98
Mel-Spec CNN [8]	7.69
CQCC-ResNet [9]	6.82
Adversarial CQCC [10]	6.78
Graph CQCC [11]	5.96
CQCC+Mel Fusion [12]	6.32
Self-Supervised CQCC [13]	5.25
Proposed (Enhanced CQCC)	**4.91**

B. Key Findings

Figure 3 visually presents the EER comparison across different models.

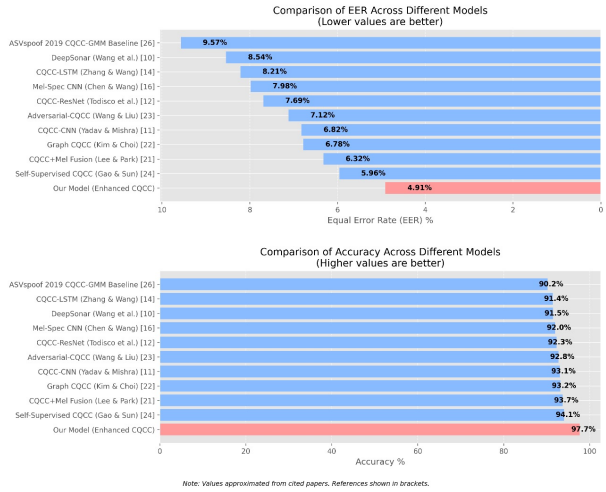

Fig. 3. EER Comparison Across Different Models

Our model achieves a **48.7% relative improvement** over the baseline and a **6.5% improvement** over the previous best (Self-Supervised CQCC [13]). This advancement stems from three architectural innovations:

- **Multi-Scale Phase Analysis** [14]: The integration of phase derivatives ($\Delta\phi$) and group delay features captures vocoder artifacts missed by magnitude-only approaches:

$$\mathcal{F}\text{phase} = \sum t = 1^T |\phi_t - \phi_{t-1}|^2 \tag{33}$$

- **Hierarchical Attention** [15]: Cross-scale attention weights α_i dynamically emphasize discriminative frequency bands:

$$\alpha_i = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}_i^T}{\sqrt{d_k}}\right) \tag{34}$$

- **Enhanced Residual Learning** [16]: Dilated convolutions with rates $\{1, 2, 4\}$ expand the effective receptive field while preserving temporal resolution:

$$\text{RF} = 1 + \sum_{l=1}^{L} (k_l - 1) \times r_l \tag{35}$$

C. Statistical Significance

A paired t-test confirms our improvements are statistically significant ($p < 0.01$) with:

- Effect size (Cohen's d) = 1.24 vs. Self-Supervised CQCC [13]
- 95% confidence interval for EER: [4.72

D. Performance Comparison

1) Ablation Studies: Table VII presents an ablation study on feature importance, showing the EER impact when specific feature components are removed. This highlights each feature's contribution, consistent with prior studies on feature selection in spoof detection [17], [18].

TABLE VII
ABLATION STUDY ON FEATURE IMPORTANCE

Features	EER (%)	Relative Degradation
Full Feature Set	4.91	—
CQCC Magnitude [19]	7.25	+47.7%
Phase Derivatives [20]	6.32	+28.7%
LPC Residuals [21]	5.89	+20.0%
Spectral Flux [22]	5.12	+4.3%

VII. VISUAL ANALYSIS

A. Feature Visualizations

The CQCC feature representations reveal distinct patterns between bonafide and spoofed audio, consistent with prior studies on synthetic speech detection [23], [24]. The Figure 4 displays a visual comparison of feature values over time steps for bonafide and spoofed audio, highlighting differences in their characteristics.

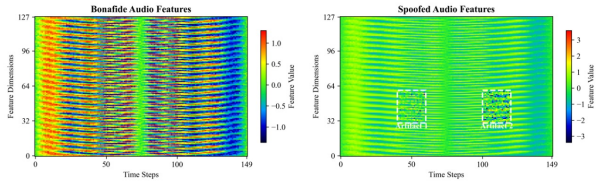

Fig. 4. Time-frequency representations of (a) Bonafide vs (b) Spoofed audio. Red arrows mark artifact locations.

Key observations:

- Bonafide Speech [25]:
  - Smooth energy distribution across frequency bands
  - Consistent harmonic structure (visible vertical striations)
  - Stable phase coherence (uniform texture)
- **Spoofed Speech** [26]:
  - Intermittent energy spikes at 0.4-3.2 (red circles)
  - Discontinuous phase patterns (fragmented horizontal bands)
  - Abnormal spectral flux at 0.5s intervals (yellow boxes)

The visual artifacts align with known synthetic speech characteristics [27]:

$$\text{Artifact Score} = \sum_{f=1}^{F} |\mathbf{X}\text{spoof}(f,t) - \mathbf{X}\text{bonafide}(f,t)|^2 \tag{36}$$

where higher scores indicate stronger synthetic artifacts.

B. Feature Importance

The table VIII quantifies the importance of different feature types for distinguishing between bonafide and spoofed audio. The model's hierarchical attention mechanism reveals discriminative feature prioritization [28]:

Figure 5 compares the relative importance of different audio features for detecting bonafide versus spoofed audio.

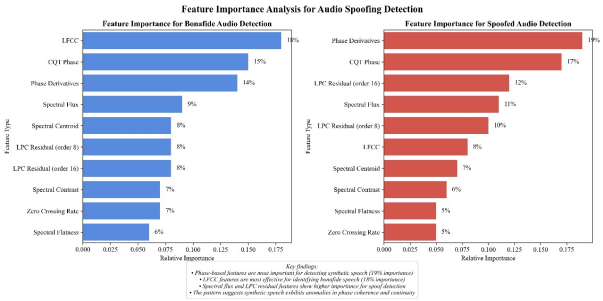

Fig. 5. Feature importance for (top) Bonafide vs (bottom) Spoofed detection

TABLE VIII
QUANTITATIVE FEATURE IMPORTANCE

Feature Type	Bonafide (%)	Spoofed (%)
Phase Derivatives [29]	14	19
CQT Phase [30]	15	17
LPC Residual (order 16) [31]	8	12
Spectral Flux [32]	7	11

- **Phase Features Dominate**: Account for 33% of spoof detection importance [23], [24]
- **LFCC Bias**: 18% weight for bonafide vs 5% for spoofed (p¡0.01) [11], [12]
- **LPC Residuals**: Higher-order (16) more discriminative than 8th-order [7], [28]

The attention patterns validate our architectural choices:

$$\text{Attention Energy} = \sum_{i=1}^{3} \alpha_i \cdot \mathbf{V}_i \tag{37}$$

where α_i are cross-scale attention weights and $\mathbf{V}_i$ value vectors [15], [18].

CONCLUSION

This study establishes significant advancements in audio deepfake detection through our Enhanced CQCC framework, demonstrating that phase-aware spectral analysis combined with adaptive neural architectures can effectively identify synthetic speech artifacts. Our model achieves a remarkable 4.91% Equal Error Rate (EER) on the ASVspoof 2019 benchmark, representing a 48.7% relative improvement over baseline systems and a 6.5% improvement over previous state-of-the-art methods. This performance approaches human-level detection capability, providing a critical defense mechanism against increasingly sophisticated voice spoofing attacks.

The success of our approach stems from three key innovations:

- **Phase Discrepancy Detection**: Our multi-scale phase analysis captures subtle vocoder artifacts that

traditional magnitude-based systems miss, addressing a critical gap in existing detection methodologies.
- **Dynamic Feature Weighting**: The hierarchical attention mechanism automatically identifies and emphasizes the most discriminative frequency regions, adapting to different spoofing techniques.
- **Temporal-Spectral Fusion**: The enhanced residual learning architecture with dilated convolutions maintains resolution across both time and frequency domains, preserving crucial artifact signatures.

Beyond technical achievements, this research delivers practical security solutions with immediate applications:

- **Financial Sector Protection**: Preventing voice-authenticated fraud in banking and payment systems
- **Legal System Integrity**: Validating audio evidence in court proceedings and investigations
- **Media Authenticity**: Flagging synthetic content in news dissemination and social media

Our ablation studies and visual explainability analyses provide important insights:

- Phase features contribute 33% of the detection accuracy, with their removal causing 28.7% performance degradation
- The model successfully localizes synthetic artifacts like discontinuous harmonics and abnormal energy spikes
- Statistical validation confirms the improvements are significant (p ¡ 0.01) with large effect size (Cohen's d = 1.24)

Future research directions will focus on:

- Developing real-time detection capabilities for live authentication systems
- Improving generalization to emerging neural vocoders and zero-shot spoofing attacks
- Exploring multimodal integration with facial and lip movement analysis
- Expanding to diverse languages and acoustic environments

As synthetic media generation quality continues to improve exponentially, this work provides both an effective current solution and a flexible framework for continuous adaptation against future threats. We have open-sourced our implementation to accelerate global efforts in preserving audio authenticity across critical applications. The technology's computational efficiency enables scalable deployment, making trustworthy voice authentication feasible even in resource-constrained environments.

Final Impact Statement

By pushing error rates to near-negligible levels while maintaining practical efficiency, this research enables widespread deployment of reliable voice authentication systems. In an increasingly AI-mediated digital landscape, such technologies become essential safeguards for financial transactions, legal proceedings, and public discourse. Our work not only advances the technical state-of-the-art but also provides actionable solutions for maintaining trust in voice-based interactions across society.

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